

The Application Layer: Network programming and DNS

Lecture 9



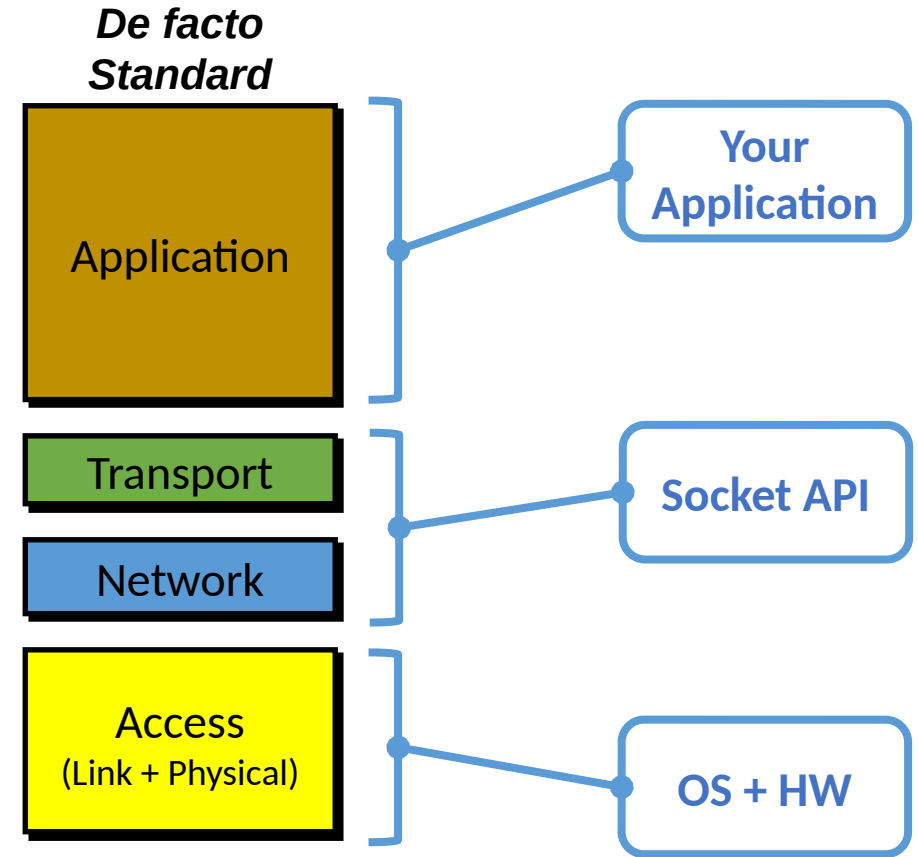
UNIVERSITÀ DEGLI STUDI DI NAPOLI
FEDERICO II



Application Layer

Creating Network Applications

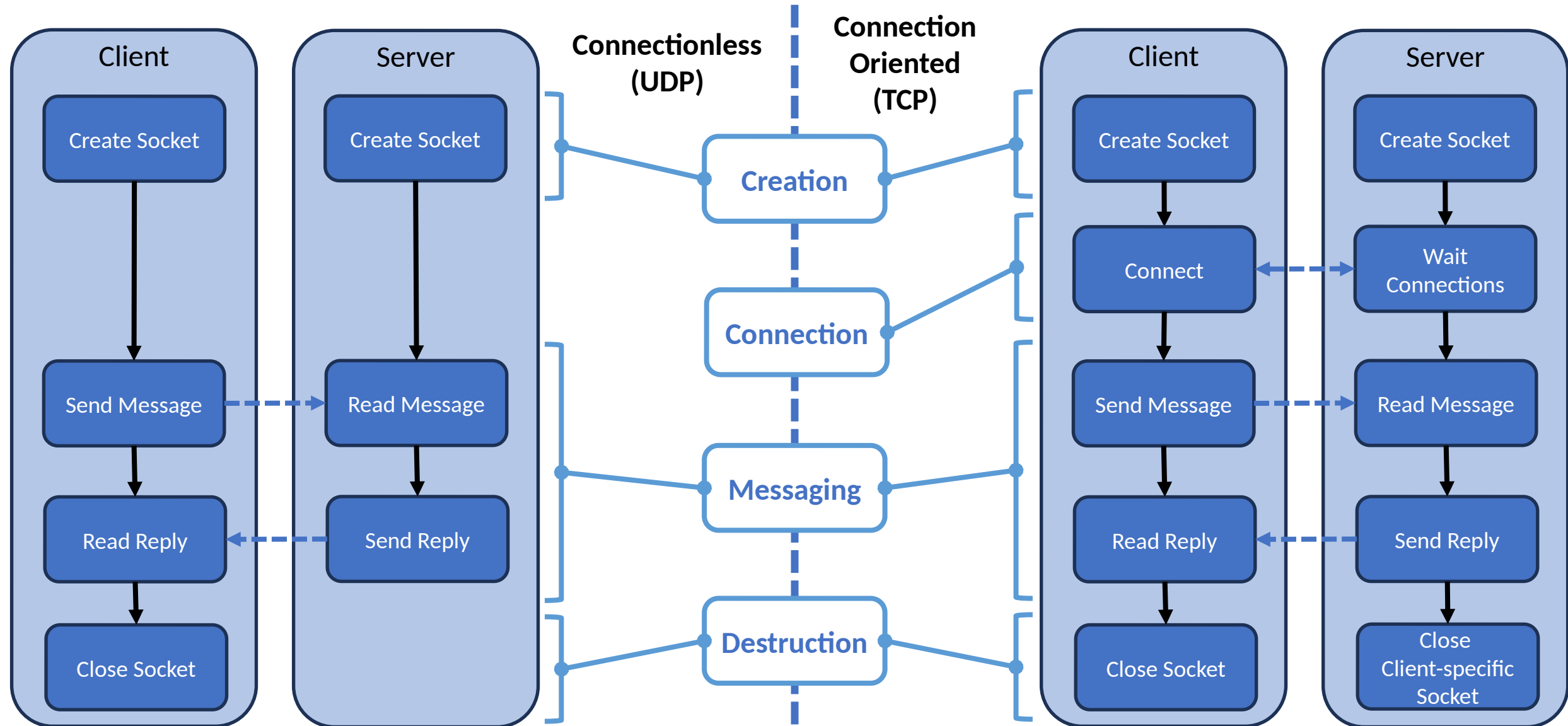
- **Sockets** are a central element for the creation of network applications.
- They typically manages (black-box) the transport (UDP/TCP) and the network (IP) layers functionalities.
- Note that there are several applications (middleware) that may **wrap sockets** simplifying their creation and providing more complex functionalities.
- However, **basic API are still widely used.**



Where layers are implemented

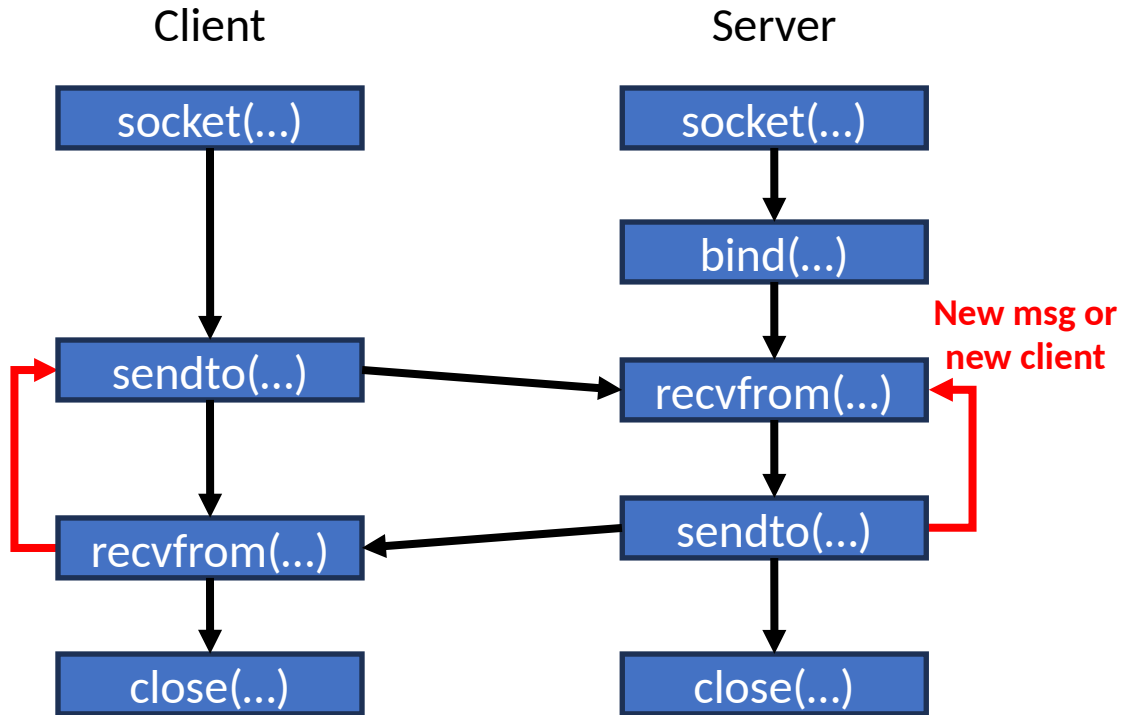
Application Layer

Connection-oriented vs. Connectionless



Application Layer

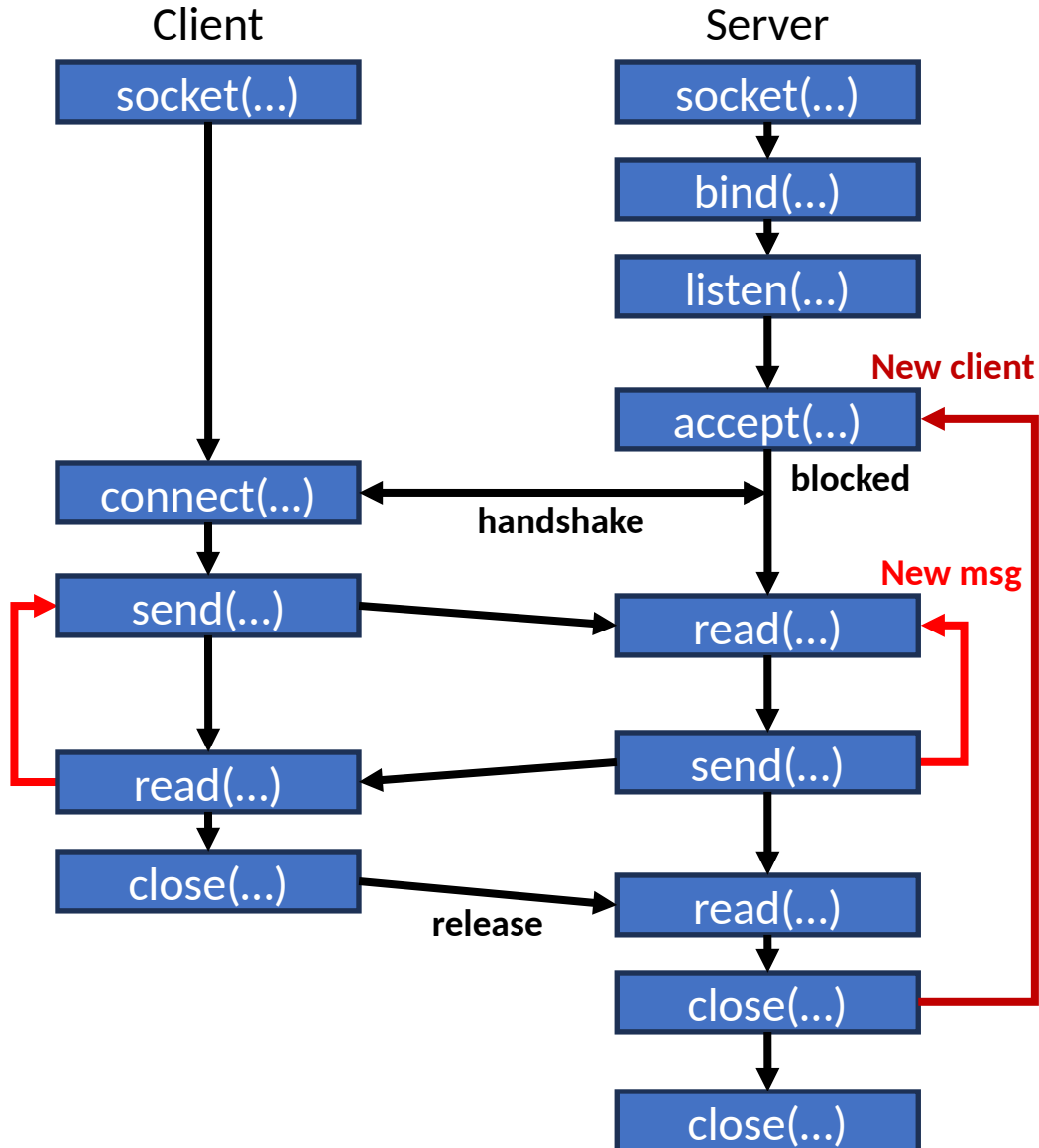
UDP Socket Commands



- For UDP connection we need both client and server to open a socket.
- The server has to bind the socket to a specific port.
- Then we can exchange messages using `sendto` and `recvfrom` functions.
 - Here IP and port of the client are retrieved through the `recvfrom` function.
 - We may loop over `sendto/recvfrom` to keep alive the communication.
- When communication is over, both hosts close the socket.
 - Notice that if client only closes the socket, the server may accept other clients.

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TCP Socket Commands



- For TCP connection we need both client and server to open a socket.
- The server has to bind the socket to a specific port, listen and eventually accept new connection. The client has to connect to the server.
- Then we can exchange messages using send and read functions.
 - Here IP and port of the client are retrieved from the newly created socket.
 - We may loop over send/read to keep alive the communication.
- When communication is over, both hosts close the socket.
 - Notice that the server should perform an additional read before to close the client-specific socket (connection release) otherwise we must wait about 4 minutes to reuse the port.
- The server may also go back to the accept function and welcome another client.

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DNS Translation (C/C++)

- As just shown, we can create sockets between two hosts, but **we need to know (IP) address and port of the server.**
- On the Internet, hosts are typically identified by **hostnames rather than by IP addresses.**
- Berkeley sockets (C/C++) only works with IP addresses.
 - This is not surprising as **sockets works on transport layer**, while **hostnames works on application layer.**
- If we are to connect with a host by hostname, **we need to use DNS translation** and get the associated IP address to be passed to the socket.

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DNS Translation (C/C++)

- C/C++ offers some functions and structures that perform such translation for us, which are used **to contact DNS servers** and to retrieve addresses.
- In particular, we can use the function *gethostbyname()* to contact DNS servers.
- The above function **does not return just the IP address**. It returns a *hostent* structure that contains some info about the host:
 - Canonical name.
 - Possible aliases.
 - One or more addresses.

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DNS Translation (C/C++)

- The hostent structure is defined as follows:

```
#include <netdb.h>

struct hostent {
    char * h_name;           // original name of the host (canonical)
    char ** h_aliases;       // list of aliases (terminated by a NULL pointer)
    int  h_addrtype;         // family of the address (typically AF_INET)
    int  h_length;           // length of the address (typically 4 bytes)
    char ** h_addr_list;     // list of addresses (terminated by a NULL pointer)
};

// for compatibility reasons
#define h_addr h_addr_list[0]
```

- A DNS may provide a list of addresses (which may be randomly ordered if the query is sent to round-robin DNS servers).
 - We can get the address of the first element of the array (`h_addr_list[0]` or `h_addr`).
- The returned addresses can be casted to *in_addr* structure used by sockets.

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DNS Translation (C/C++)

- The `gethostbyname()` function is defined as follows:

```
#include <netdb.h>

struct hostent *host_info = gethostbyname(const char *name);
```

- Where:
 - `name`: is a string containing the hostname to be translated.
 - `host_info`: is a pointer to the `hostent` structure containing the information about host.
 - This is `NULL` if error occurred.
- Notice that there is also a similar function *`gethostbyaddr()`* that returns a `hostent` structure for a specified IP address.

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DNS Translation (C/C++)

```
#include <iostream>
#include <cstring>
#include <string>
#include <sys/socket.h>
#include <arpa/inet.h>
#include <sys/types.h>
#include <netinet/in.h>
#include <unistd.h>
#include <netdb.h>
#include <stdio.h>
```

```
int main(int argc, char **argv){
    struct hostent *server_info;
    char *server_name;

    if(argc < 2){
        std::cout<<"No hostname specified"<<std::endl;
        return 0;
    }
    else {
        server_name = argv[1];
    }

    server_info = gethostbyname(server_name);
    if (server_name == NULL){
        std::cout << "could Not resolve hostname " << server_name << " :(" << std::endl;
        return 0;
    }
}
```

- Include libraries to allow DNS translation.
- Initialization.
- Get hostname as argument.
- Invoke DNS and check if a response has been received.

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DNS Translation (C/C++)

```
//plot output
std::cout<<"Canonical name:"<<std::endl;
std::cout<<"\t"<<server_info->h_name<<std::endl;

std::cout<<"Aliases:"<<std::endl;
int i = 0;
while(server_info->h_aliases[i] != NULL){
    std::cout<<"\t"<<server_info->h_aliases[i]<<std::endl;
    i++;
}

std::cout<<"IPs:"<<std::endl;
i = 0;
while(server_info->h_addr_list[i] != NULL){
    std::cout<<"\t"<< inet_ntoa( *( (struct in_addr *) server_info->h_addr_list[i] ) ) <<std::endl;
    i++;
}

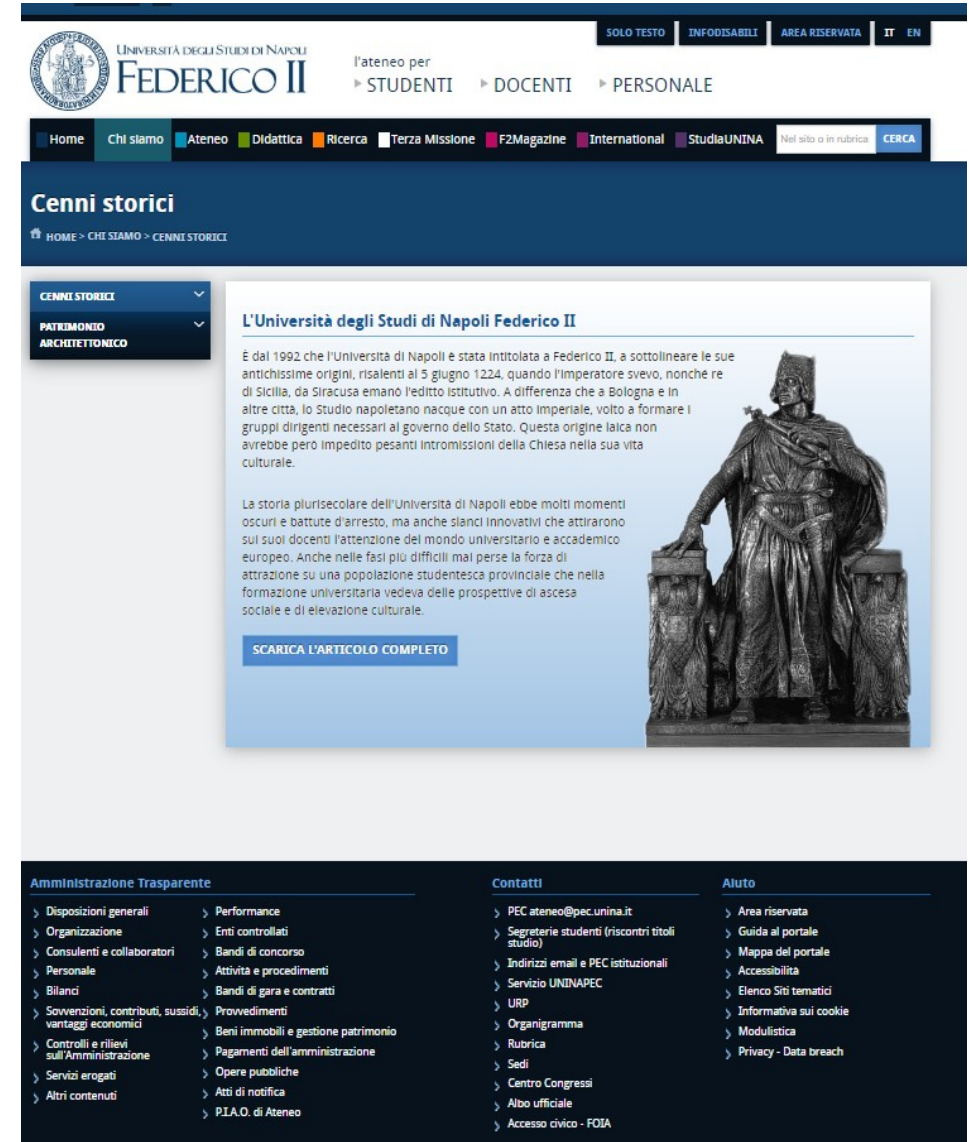
return 0;
}
```

- Plotting output from hostent structure:
 - Canonical name.
 - List of aliases
 - List of IPs.

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HTTP Socket Example

- We will now create a C++ code implementing from scratch a HTTP request (HEAD) for a web page.
- The web page we will check is the “cenni storici” from the UNINA website:
 - <http://www.unina.it/chi-siamo/cenni-storici>



Application Layer

HTTP Socket Example (C/C++)

```
#include <iostream>
#include <cstring>
#include <string>
#include <sys/socket.h>
#include <arpa/inet.h>
#include <sys/types.h>
#include <netinet/in.h>
#include <unistd.h>
#include <netdb.h>
#include <stdio.h>
```

```
int main(){
    int socket_desc;
    struct sockaddr_in serv_addr;
    struct hostent *server;
    char buffer[4096];

    socket_desc = socket(AF_INET, SOCK_STREAM, 0);
    if (socket_desc < 0){
        std::cout << "failed to create socket" << std::endl;
        return 0;
    }
}
```

- Include libraries to allow TCP connection to the web server.
- Initialize variables and create the TCP socket. For a HTTP request 3 elements are specified:
 - Hostname of the web server.
 - Port number.
 - Resource to be retrieved (object's path).
- Note: here we use the C++ cout for simplicity.

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HTTP Socket Example (C/C++)

```
server = gethostbyname("www.unina.it");
if (server == NULL){
    std::cout << "could Not resolve hostname :("
        << std::endl;
    close(socket_desc);
    return 0;
}

bzero((char *) &serv_addr, sizeof(serv_addr));
serv_addr.sin_family = AF_INET;
serv_addr.sin_port = htons(80);
bcopy((char *)server->h_addr, (char *)&serv_addr.sin_addr.s_addr, server->h_length);

if (connect(socket_desc, (struct sockaddr *)&serv_addr, sizeof(serv_addr)) < 0){
    std::cout << "connection failed :(" << std::endl;
    close(socket_desc);
    return 0;
}
```

- Connection to the web server. Here we use the following function:

```
server =
gethostbyname(host.c_str());
```

that invokes DNS and to gets the IP of the server from the hostname (inside the *hostent* struct).

- The IP from the returned structure is copied into the *serv_addr* structure for the following connect function.

Application Layer

HTTP Socket Example (C/C++)

```
const char *request = "HEAD /chi-siamo/cenni-storici HTTP/1.1\r\nHost: www.unina.it\r\nConnection: close\r\n\r\n";

if (send(socket_desc, request, strlen(request), 0) < 0){
    std::cout << "failed to send request..." << std::endl;
    close(socket_desc);
    return 0;
}
std::cout<<"message sent:"<<std::endl;
std::cout<<request<<std::endl;

int n = recv(socket_desc, buffer, sizeof(buffer), 0);
```

```
std::cout<<"received "<<n<<" bytes:"<<std::endl;
std::cout<<buffer;

close(socket_desc);

return 0;
```

- Messaging with the web server. The HTTP request is defined into the request string.
- Here we are creating a non-persistent connection since we just want to check one page.
- Close socket and print the received page (HTML).